



# Poland — Substation ESG Report

v4.2

Substation-level Environmental, Social, and Governance assessment aligned to CSRD/ESRS, EU Taxonomy, TCFD, SFDR, and UN PRI frameworks, across the 2,248-substation Poland fleet. Updated monthly via the SSI Index v4.2 scoring engine (101 variables · 34 sources · 11 modifiers · Re composite bounded [0.920, 1.787]).

SUBSTATION

PL523 · Station  
v43\_ff22713eca7d

PL\_v43\_ff22713eca7d · PL523,

SSI SCORE (R<sub>MEDIAN</sub>)

0.094

● Low Risk 18.0th percentile

VOLTAGE CLASS

0 kV

distribution-tier

CONFIDENCE INTERVAL

0.094 – 0.094

P5–P95 · CI width 0.000 · high confidence

MARKOV ETTC

— yr

Expected time to criticality

DER PENETRATION

—

T1 transition stress score

## Component Fingerprint

+ Re composite v4.2

Six-component weighted decomposition of the SSI composite score (C / V / I / E / S / T) — the substation's structural risk profile — PLUS the v4.2 **Re composite**, a bounded resilience aggregate of the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just). Layout: *left* — the 6 components rendered as a weighted contribution bar · *right* — the same 6 components rendered as horizontal bars side-by-side with Re\_norm as the comparable 7th bar (terracotta) on the [0, 1] scale · *below* — the Re composite tier card (numeric value + bounded trajectory bar + classification + formula explainer). Re composite re-appears as the 7th axis of the ESG Radar in Section A below.

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WEIGHTED CONTRIBUTION TO  $R_{BASE}$

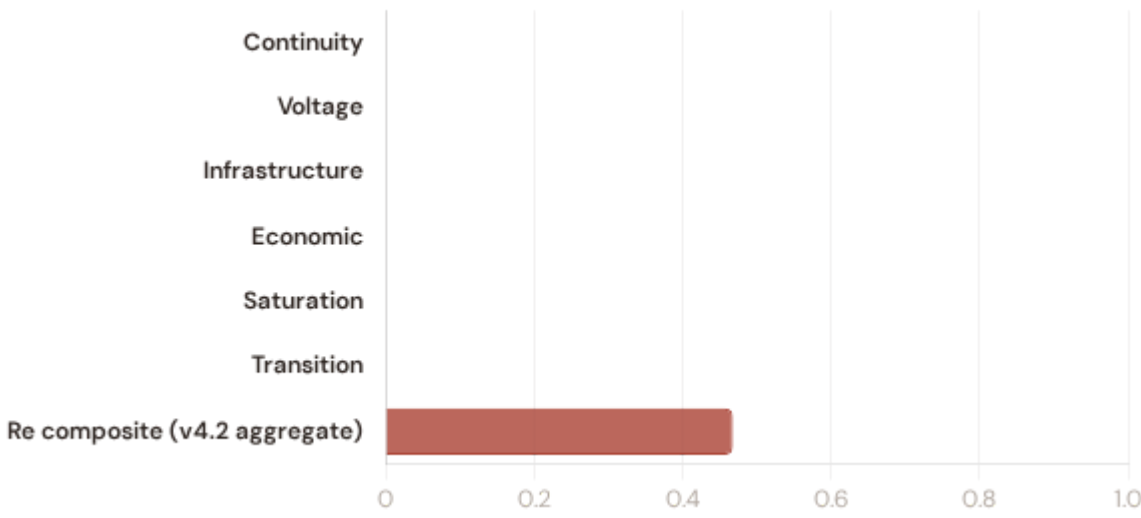


V4.2 RE COMPOSITE (AGGREGATE, NOT IN  $R_{BASE}$ ) moderate exposure



- Continuity —  $\times 0.3$  ■ Voltage —  $\times 0.1$  ■ Infrastructure —  $\times 0.25$  ■ Economic —  $\times 0.1$
- Saturation —  $\times 0.2$  ■ Transition —  $\times 0.05$  ■ Re composite v4.2 aggregate **0.468** agg

■  $R_{base}$  total (weighted sum) **0.000**  $\Sigma = 1.00$



**0.468**

RE\_NORM

**MODERATE EXPOSURE**

LOW MODERATE ELEVATED (1.00)  
(0.00) (0.25) (0.50)



**Re\_raw** 1.3256 (bounds [0.920, 1.787]; normalised to [0, 1])

FORMULA

$$Re\_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$$

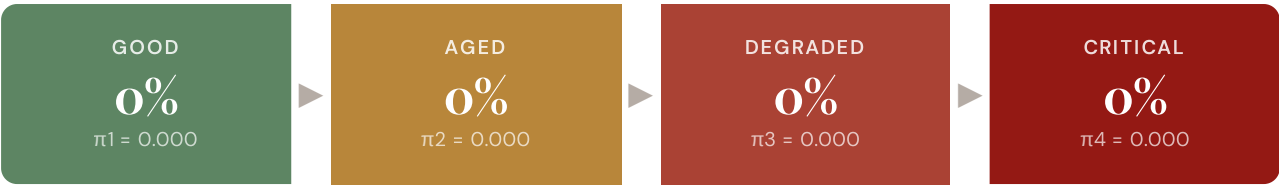
Bounded resilience aggregate of the v4.2 family.  $Re < 1.0$  when R8 adaptive capacity dominates other near-neutral factors.

Re composite tier classification: **low** ( $Re\_norm < 0.25$ ) · **moderate** ( $0.25-0.50$ ) · **elevated** ( $\geq 0.50$ ). Bounds [0.920, 1.787] map to  $Re\_norm$  [0, 1] via  $clip((Re\_raw - 0.920) / (1.787 - 0.920), 0, 1)$ .

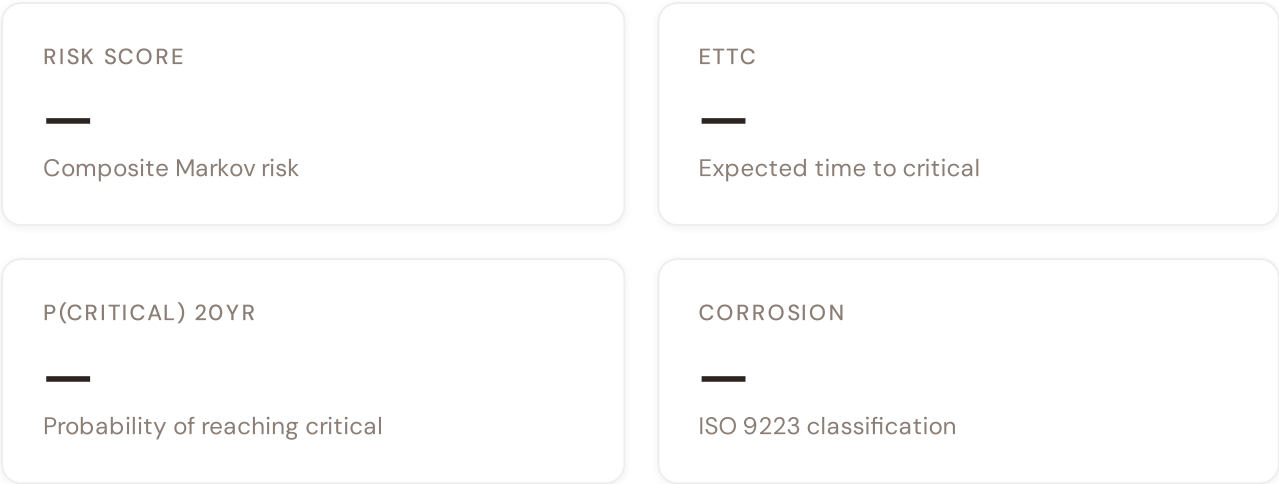
## Markov Degradation Profile

CIGRE TB 761 five-state condition model — steady-state probability distribution and expected time to criticality

STEADY-STATE DISTRIBUTION

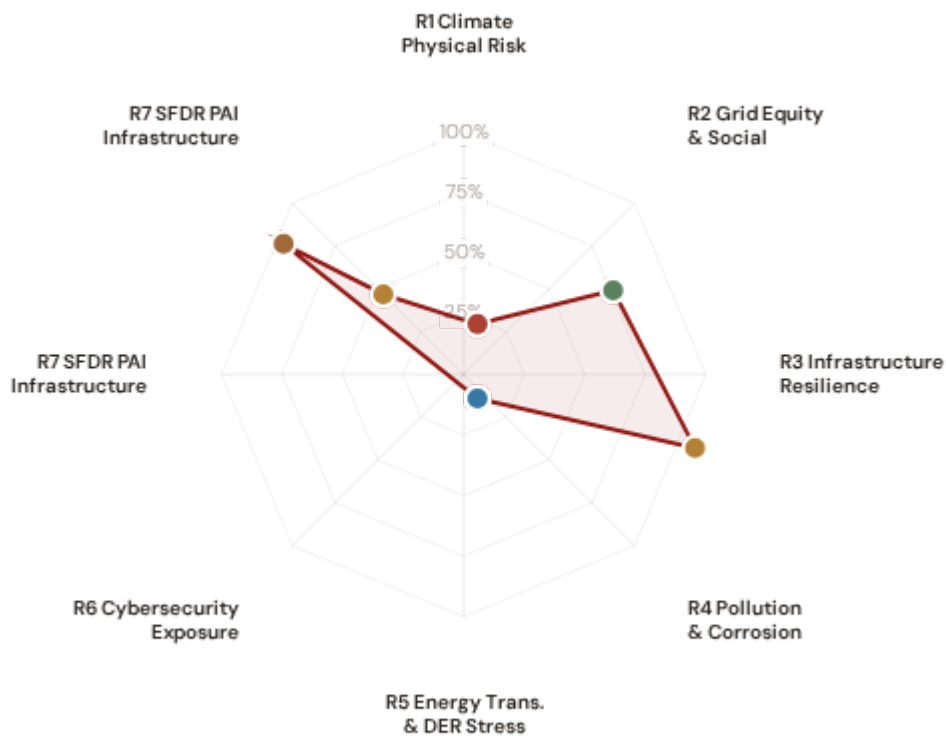


Markov 5-state model calibrated from CIGRE Technical Brochure 761 (2019). States represent asset condition from new/good through critical. Steady-state probabilities indicate the long-run proportion of time the asset spends in each condition state given current operating environment.



ESG Radar — Seven-Report Overview <sup>v4.2</sup>

A Composite readiness across 7 ESG dimensions per FC v3 §14 (R1 Climate Physical · R2 Grid Equity & Social · R3 Infrastructure Resilience [Re composite ESG home] · R4 Pollution & Corrosion · R5 Energy Transition & DER Stress · R6 Cybersecurity · R7 SFDR PAI Infrastructure, FC v3 §14 subsection 13.7), each linked to UN Sustainable Development Goals. *Note: R7 here is the SFDR ESG-disclosure axis (Re\_norm proxy under Convention #7) — distinct from the v4.0.2 formula-chain R7 cyber/digital modifier documented in methodology.html. The two R7s share a number by design (V4\_2\_IMPLEMENTATION\_ARCHITECTURE.md §4.5).*



Re composite v4.2 7th axis · Re\_raw 1.3256 · Re\_norm 0.468 · MODERATE EXPOSURE



### R1 · Climate Physical Risk Assessment GAP

SDG 13 Climate Action · SDG 9 · SDG 11

SDG 13

SDG 9

SDG 11

### R2 · Grid Equity & Social Vulnerability PARTIAL

SDG 7 Affordable and Clean Energy · SDG 10 · SDG 1

SDG 7

SDG 10

SDG 1

### R3 · Infrastructure Resilience [Re composite home] READY

SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 12

SDG 9

SDG 13

SDG 12

### R4 · Pollution & Corrosion GAP

SDG 11 Sustainable Cities and Communities · SDG 3 · SDG 15

SDG 11

SDG 3

SDG 15

### R5 · Energy Transition & DER Stress GAP

SDG 7 Affordable and Clean Energy · SDG 13 · SDG 9

SDG 7

SDG 13

SDG 9

**R6 · Cybersecurity Exposure** **GAP**

SDG 9 Industry, Innovation, Infrastructure · SDG 16

SDG 9

SDG 16

**R7 · SFDR PAI Infrastructure Disclosure** **READY**

SDG 12 Responsible Consumption and Production · SDG 9 · SDG 13

SDG 12

SDG 9

SDG 13

**R7 · SFDR PAI Infrastructure** **DEFAULT** **v4.2**

SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 11 · Re\_norm = 0.468 (moderate exposure)

SDG 9

SDG 13

SDG 11

# 1 Climate Physical Risk Assessment

ESRS E1 · TCFD Physical Risk · EU Taxonomy Annex A

**GAP****WHY THIS REPORT EXISTS**

ESRS E1 (Climate Change) is material for 98% of CSRD-reporting companies. TCFD physical risk disclosure is mandatory in 36+ jurisdictions. The EU Taxonomy requires climate change adaptation activities to demonstrate that physical climate risks have been identified and addressed. This report quantifies acute and chronic climate hazard exposure at substation level using ERA5 reanalysis and Markov degradation modelling.

**SDG Alignment****SDG 13 — Climate Action** (primary)

Target 13.1 calls for strengthening resilience and adaptive capacity to climate-related hazards. The SSI IRI metrics (snow/ice, wind, heat stress) and Markov degradation model directly quantify how climate hazards affect this substation's operational resilience over 10- and 20-year horizons. With a Markov risk score of — and ETTC of — years, this substation's climate trajectory can be assessed against TCFD physical risk and EU Taxonomy adaptation screening requirements.

SDG 13

SDG 9

SDG 11

### Substation Profile

|                               |                  |
|-------------------------------|------------------|
| Insulation Risk (I component) | —                |
| Environment Component (E)     | —                |
| Seismic PGA                   | 0.0179g (Zone 4) |
| Markov Risk Score             | —                |
| ETTC (time to criticality)    | — years          |
| P(critical) at 20 years       | —                |
| Corrosion Class               | —                |
| Climate Trajectory (R2)       | NOT ACTIVE       |

### SSI Variables Feeding This Report

| SSI VARIABLE                    | VALUE   | ESG DISCLOSURE REQUIREMENT                       | STATUS |
|---------------------------------|---------|--------------------------------------------------|--------|
| I1 Snow/Ice IRI                 | 0.900   | ESRS E1-9: Financial effects from physical risks | READY  |
| I2 Tree-fall IRI                | 1.050   | TCFD Strategy (b): Physical risk impact          | READY  |
| I3 Heat-wave IRI                | 1.150   | EU Taxonomy Annex A: Heat stress                 | READY  |
| I5 Thermal stress (I aggregate) | —       | ESRS E1-4: Climate mitigation targets            | GAP    |
| R2 Climate trajectory           | N/A     | TCFD Strategy (c): Scenario analysis             | GAP    |
| R6b Seismic PGA                 | 0.0179g | EU Taxonomy Annex A: Geophysical hazards         | READY  |
| Markov p_critical_10yr          | —       | TCFD Risk Management: Risk identification        | READY  |
| Markov p_critical_20yr          | —       | ESRS E1-9: Time horizons                         | READY  |

## Data Sources & Vintage

ERA5 Climate Reanalysis

Copernicus CDS

2024 Weekly

National Seismic Hazard Map

National Geological Survey

2023 Multi-year

IEEE C57.91 Thermal Model

IEEE

Standard N/A

CIGRE TB 761 Markov

CIGRE

2019 N/A

CMIP6 SSP2-4.5 Projections

Copernicus CDS

2024 v4.2 5-model ensemble

### TECHNICAL VALIDITY

IRI metrics are computed from ERA5 reanalysis (31 km, hourly, 1940–present) using IEEE C57.91 thermal loading curves. Markov 5-state degradation is calibrated from CIGRE Technical Brochure 761 (2019) condition assessment data. Forward-looking projections use the CMIP6 SSP2-4.5 5-model ensemble (ACCESS-CM2, CNRM-CM6-1, EC-Earth3, GFDL-ESM4, MRI-ESM2-O), baseline 2000–2020 vs future 2030–2050, feeding the substation-level climate\_trajectory (I1/I2/I3) fields for 32 of 39 countries. Any PARTIAL status now reflects country-specific gaps: 7 Wave 2–3 countries (Australia, Chile, Colombia, Costa Rica, Iceland, Ireland, Israel) await the next L2 climate pass; 8 Wave 4 countries emit fleet-uniform trajectories pending the per-substation bilinear interpolation regression fix (methodology follow-on, sister to R3\_C\_mult uniformity). All other inputs are production-grade with institutional provenance.

## 2 Grid Equity & Social Vulnerability

ESRS S1/S2 · SFDR PAI Social Indicators · UN PRI

PARTIAL

### WHY THIS REPORT EXISTS

ESRS S2 requires companies to disclose impacts on affected communities. SFDR mandates 5 social PAI indicators. The SSI Index is uniquely positioned here because no competing grid risk framework integrates socio-economic vulnerability at substation resolution. This report surfaces where infrastructure risk concentrates in economically disadvantaged populations.

SDG Alignment

SDG 7 — Affordable and Clean Energy (primary)

Target 7.1 requires universal access to affordable, reliable energy. This substation serves a catchment with V\_socio of 0.54 and energy poverty rate of 10%. The R3 social modifier (—) amplifies risk scores in regions with high unemployment, elderly concentration, and net outward migration — making spatial inequality visible and quantifiable at asset level.

SDG 7

SDG 10

SDG 1

Substation Profile

|                                |          |
|--------------------------------|----------|
| V_socio (energy poverty index) | 0.54     |
| EP Rate (energy poverty %)     | 10%      |
| Unemployment Rate              | 3.4%     |
| GDP per Capita                 | € 12,500 |
| R&D % of GDP                   | 2.1%     |
| R3 Social Multiplier           | —        |
| Elderly Vulnerability          | 18.5%    |
| Migration Score                | 0.50     |

### SSI Variables Feeding This Report

| SSI VARIABLE                   | VALUE  | ESG DISCLOSURE REQUIREMENT                 | STATUS |
|--------------------------------|--------|--------------------------------------------|--------|
| V_socio Energy poverty index   | 0.54   | ESRS S2-4: Impacts on affected communities | READY  |
| EP_rate Energy poverty rate    | 10%    | SFDR PAI #14: Fossil fuel exposure proxy   | READY  |
| R3 Social multiplier           | —      | ESRS S1-6: Workforce characteristics       | GAP    |
| Elderly Elderly vulnerability  | 18.5%  | SFDR PAI: Vulnerable populations           | READY  |
| Community Catchment population | 23,406 | ESRS S2: Community impact                  | READY  |

### Data Sources & Vintage

|                                                                                                                                         |                      |
|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>GHSL Population Grid</b><br>EC JRC / Copernicus                                                                                      | 2025 Multi-year      |
| <b>Global Net-Migration Grid</b><br>Niva et al. 2023 (Aalto / Wittgenstein)                                                             | 2000–2019 Multi-year |
| <b>Eurostat GISCO NUTS-3 Regions</b><br>European Commission / Eurostat                                                                  | 2024 Annual          |
| <b>GADM Administrative Areas 4.1 (CO / CH / CA)</b><br>UC Davis <a href="https://geodata.ucdavis.edu/gadm">geodata.ucdavis.edu/gadm</a> | 2022 Multi-year      |
| <b>Economic Statistics (GDP, unemployment)</b><br>National Statistics Office                                                            | 2023 Annual          |
| <b>Energy Market Data</b><br>National TSO                                                                                               | 2023 Annual          |

#### TECHNICAL VALIDITY

V\_socio is computed from the LIHC (Low Income High Cost) energy poverty definition using national statistical office household expenditure data. The R3 modifier amplifies infrastructure risk scores in catchments with high unemployment, elderly concentration, and net outward migration. Coverage status (21 July 2026 audit): V\_socio + EP\_rate + gdp\_per\_capita + unemployment\_rate are populated for 32/39 countries, but 8 large countries (Italy, Japan, Portugal, Spain, Sweden, Germany, France, US) emit fleet-uniform national aggregates instead of per-substation LAU-2/NUTS-3 values — pending the systemic Wave 4 per-substation interpolation regression fix (sister to R3\_C\_mult + CMIP6 climate\_trajectory). (Task #452 already closed this gap for migration\_score specifically via per-substation Niva 2023 raster enrichment; the remaining V\_socio / EP\_rate / GDP / unemployment fields await Task #450 SYSTEMIC scope.) Catchment

population is now populated for all 39/39 countries via the GHSL Population Grid (EC JRC / Copernicus Emergency Management Service, R2023A epoch E2025, Mollweide equal-area) computing 5 km per-substation zonal sums — Task #451 landed 23 July 2026, replacing a legacy synthetic generator with a real GHSL spatial-join pipeline (795k-substation cohort; 0.17% receive Convention #56 visibly-honest None fallbacks for remote/offshore points where the raster has NoData — concentrated in US Aleutians+Pacific, Canada Arctic, Australia outback, Nordic fjords). Migration\_score is now populated for all 39/39 countries via the Niva et al. 2023 gridded net-migration dataset (Aalto University / Wittgenstein Centre, published Nature Human Behaviour 7:2023–2037, DOI 10.1038/s41562-023-01689-4, dataset DOI 10.5281/zenodo.7997134, CC BY 4.0 licence) using the 20-yr sum raster 2000–2019 at 5 arc-min (~10 km) resolution — Task #452 landed 23 July 2026, closing the R2 defect via NARROW-scope enrichment (fill Nones) plus fleet-uniform override that retired the 0.5 national scalar fallback of 8 Wave 4 majors (France, Germany, US, Italy, Spain, Portugal, Sweden, Japan, Australia partial, Austria partial) with real per-substation Niva-derived values in [0, 1] range via  $\tanh(x/200)$  mapping. Genuine pre-Task-#452 per-substation distributions (Norway, Denmark, Lithuania, Ireland, Turkey, Mexico, Greece, New Zealand, Poland partial) are preserved untouched — their semantic-drift issues (e.g. Mexico [-5, +8] percent-scale, Denmark tiny [-0.02, +0.04] scale) are deferred to Task #450 SYSTEMIC scope. Cohort-wide 97.13% real Niva values written + 0.17% Convention #56 visibly-honest None + 2.70% preserved distributions untouched. The remaining 4 countries with major partial-coverage gaps from failed spatial joins (Luxembourg 12% populated / Slovenia 16% / Colombia 51% / Lithuania 50%) are now closed via Task #453 (23 July 2026) using per-substation polygon spatial-join against Eurostat GISCO NUTS-3 2024 shapefile (European Commission / Eurostat, EPSG:4326, annual vintage, open-license attribution-required — for LU/SI/LT) plus GADM 4.1 admin1 shapefile for Colombia (UC Davis geodata.ucdavis.edu/gadm, peer-reviewed academic-open-license derivative of DANE Marco Geoestadístico Nacional — chosen as documented-proxy after DANE geoportal empirically requires form-based interactive download). Task #453 empirical closure: 6,967 v43 substations enriched across LU/SI/LT/CO (LU 634/634 = 100% · SI 1,571/1,574 = 99.8% · LT 4,396/4,396 = 100% · CO 366/366 = 100%; 3 Convention #56 fallback total = 0.04% cohort-wide; 100% preservation of Task #451 catchment\_population + Task #452 migration\_score audit-trail markers via merge-not-replace BINDING contract). Each Task #453 enriched substation carries per-region  $\text{gdp\_per\_capita} + \text{unemployment\_rate} + \text{EP\_rate\_region} + \text{elderly\_pct} + \text{computed V\_socio}$  (via mirrored  $0.45 \cdot \text{ep\_norm} + 0.35 \cdot \text{gdp\_norm} + 0.20 \cdot \text{elderly\_norm}$  formula from socioeconomic.py:511–517 canonical). Task #453 is the ADMIN structural variant within REPORTS\_FRAMING\_KB.md §8bis Discipline #47 spatial-enrichment family (STOCK = Task #451 GHSL zonal-sum, FLOW = Task #452 Niva point-sampling, ADMIN = Task #453 polygon spatial-join). Discovery motivating Task #453 also codified METHODOLOGY\_DISCIPLINES.md §5septies (empirical OSM tag density is per-country) as candidate architectural discipline — Poland + Luxembourg + Slovenia + Colombia + Lithuania empirically ALL show 0.0% populated ref:nuts:3 / addr:department tag rates at country scale, retiring the connector-tag-extraction path as load-bearing for admin-code derivation. Task #454 SYSTEMIC (24 July 2026) extended the Task #453 polygon utility to 11 additional countries with matching v43-at-partial-coverage pattern — 9 EU countries (Belgium + Czechia + Denmark + Estonia + Finland + Ireland + Latvia + Netherlands + Poland) via the same Eurostat GISCO NUTS-3 2024 shapefile already loaded for Task #453, plus 2 non-EU countries via GADM 4.1 admin1 shapefiles (Switzerland cantons + Canada provinces; same UC Davis academic-open-license documented-proxy pattern used for Task #453 Colombia). Two utility refinements landed during Task #454 execution to accommodate new failure modes surfaced empirically: (a) a 3-case skip-logic decision tree with new case-b (province populated by earlier connector pass but socio\_economic gap unfilled — Canada case; utility bypasses polygon join and uses the existing province directly as CSV lookup key); (b) Switzerland csv\_lookup\_aliases with 5 entries mapping GADM English/French canton names (Lucerne, Sankt Gallen etc.) to CSV German canonical forms (Luzern, St. Gallen). Task #454 SYSTEMIC empirical closure: 52,043 v43 substations enriched across 11 new countries (BE 5,428 + CZ 7,825 + DK 2,364 + EE 1,178 + FI 135 + IE 282 + LV 3,425 + NL 3,807 + PL 25,509 + CH 865 + CA 1,227) plus Task #453 4-country idempotent no-op preservation of 6,967 subs = 59,041/59,077 = 99.94% cohort-wide coverage across 15-country combined Task #453+#454 cohort with 65 Convention #56 out-of-polygon fallback (0.11%) and 100% Task #451/#452 marker preservation via merge-not-replace BINDING contract. Wall-clock 15s cumulative (Poland alone 1.3s = ~19,600 subs/sec). Post-Task-#454 Discipline #47 ADMIN variant instance count is 15 (Task #453 4 + Task

#454 11), and Convention #78 §5 septies OSM-tag-density empirical instance count is 16 (Poland P21 baseline + Task #453 cohort 4 + Task #454 cohort 11 — well above Convention #76 BINDING promotion threshold of 5–10 empirical instances). All inputs sourced from institutional statistical offices with citable vintage — meeting CSRD Article 29a limited assurance requirements for the 32-country cohort where fields are per-substation, plus Task #453+#454 15-country cohort now closed at Grid Equity axis via polygon spatial-join.

## 3 Infrastructure Resilience [Re composite home]

FC v3 §14 · SSI v4.2 Re composite (canonical) · EU Taxonomy Art. 11 (mapped)

READY

### WHY THIS REPORT EXISTS

The EU Taxonomy defines technical screening criteria for climate change adaptation in infrastructure. This is the lowest-hanging ESG product from the SSI Index — READY for 9 of 10 countries today. The SSI composite score (R\_median), 6 components, modifier architecture, and Markov risk outputs provide the quantitative evidence base for Article 11 screening at asset level.

### SDG Alignment

#### SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.4 calls for upgrading infrastructure to make it sustainable. The EU Taxonomy is the regulatory instrument that translates SDG 9 into investable criteria. By classifying this substation against Article 11 technical screening criteria — using R\_median (0.094), all 6 components, and 5 modifiers — this report provides asset-level Taxonomy alignment that enables sustainable finance reporting.

SDG 9

SDG 13

SDG 12

### Substation Profile

|                        |          |
|------------------------|----------|
| R_median (composite)   | 0.094    |
| R_base (pre-modifier)  | 0.000    |
| Modifier Impact        | 9400.0%% |
| C (Condition)          | —        |
| V (Voltage)            | —        |
| I (Insulation/Climate) | —        |
| E (Environment)        | —        |
| S (Seismic)            | —        |
| T (Transition)         | —        |

### SSI Variables Feeding This Report

| SSI VARIABLE             | VALUE       | ESG DISCLOSURE REQUIREMENT                    | STATUS |
|--------------------------|-------------|-----------------------------------------------|--------|
| R_median Composite score | 0.094       | EU Taxonomy Art. 11: Adaptation screening     | READY  |
| 6 Comp. C/V/I/E/S/T      | All present | EU Taxonomy TSC: Physical risk identification | READY  |
| R3-R7 Modifier suite     | All applied | ESRS E1: Adaptation measures evidence         | READY  |
| Markov Degradation model | —           | TCFD: Forward-looking risk assessment         | READY  |
| R5 Asymmetric CI         | 0.000       | ESRS: Uncertainty quantification              | GAP    |

### Data Sources & Vintage

ERA5 Climate Reanalysis  
Copernicus CDS

2024 Weekly

National Seismic Hazard Map  
National Geological Survey

2023 Multi-year

Economic Statistics (GDP, unemployment)  
National Statistics Office

2023 Annual

CIGRE TB 761 Markov  
CIGRE

2019 N/A

#### TECHNICAL VALIDITY

Article 11 screening uses the full SSI v4.0.2 scoring engine: 6-component weighted composite (C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05), Gaussian copula Monte Carlo (10,000 iterations), and 5 modifiers (R3 social, R4 topology, R5 asymmetric CI, R6a restoration, R7 cyber). The Markov 5-state degradation model provides the forward-looking element. All inputs are from institutional open-source data with citable vintage — meeting CSRD Article 29a limited assurance requirements.

## 4

### Pollution & Corrosion

ESRS E2 Pollution · ISO 9223 Corrosion Classification · Environmental Impact

GAP

#### WHY THIS REPORT EXISTS

Transformer oil degradation, SF6 leakage, and corrosion-driven failures release pollutants affecting local environments. ESRS E2 requires disclosure of pollution risks. This report assesses the substation's corrosion classification under ISO 9223 and the environmental sensitivity of its surroundings.

#### SDG Alignment

##### SDG 11 — Sustainable Cities and Communities (primary)

Target 11.6 aims to reduce the adverse environmental impact of cities. The corrosion class (—) and E2\_local enrichment factor (—) track environmental degradation and pollution sensitivity at asset level. Where corrosion is advanced and maintenance deferred, equipment failure risks releasing mineral oil, PCBs (legacy units), and SF6 — directly relevant to urban and peri-urban environmental quality.

SDG 11

SDG 3

SDG 15

### Substation Profile

|                            |   |
|----------------------------|---|
| Corrosion Class (ISO 9223) | — |
| E2 Local Enrichment        | — |
| E Component Score          | — |
| Markov Steady-State        | — |

### SSI Variables Feeding This Report

| SSI VARIABLE                                  | VALUE | ESG DISCLOSURE REQUIREMENT             | STATUS         |
|-----------------------------------------------|-------|----------------------------------------|----------------|
| <div>Corrosion</div> ISO 9223 class           | —     | ESRS E2: Pollution risk classification | <div>GAP</div> |
| <div>E2_local</div> Environmental sensitivity | —     | ESRS E2: Local environmental impact    | <div>GAP</div> |
| <div>E</div> Environment component            | —     | ISO 9223: Atmospheric corrosion        | <div>GAP</div> |

### Data Sources & Vintage

|                                                  |                         |
|--------------------------------------------------|-------------------------|
| Energy Market Data<br>National TSO               | <div>2023</div> Annual  |
| Renewable Installations<br>National DER Registry | <div>2024</div> Monthly |

#### TECHNICAL VALIDITY

Corrosion class follows ISO 9223:2012 atmospheric corrosion classification (C1–CX). Status depends on whether the corrosion class shows real variance across the fleet or uses default values. Full validation requires national environmental agency air quality monitoring overlay at substation coordinates (SO2, NOx, particulate deposition).

5

## Energy Transition & DER Stress

ESRS E1 Transition Plan · TCFD Transition Risk · EU Green Deal Alignment

GAP

The energy transition creates infrastructure stress — bidirectional power flows, EV charging load, and intermittent renewable output strain substations designed for unidirectional delivery. This report measures whether grid infrastructure is keeping pace with decarbonisation.

SDG Alignment

SDG 7 — Affordable and Clean Energy (primary)

Target 7.2 requires substantially increasing the share of renewable energy. This substation has a DER ratio of —. The T1\_score (—) measures the stress this places on the substation, providing the empirical feedback loop between SDG 7 ambition and infrastructure reality.

SDG 7

SDG 13

SDG 9

Substation Profile

|                              |      |
|------------------------------|------|
| T1 Score (transition stress) | —    |
| DER Ratio                    | —    |
| DER Variability              | —    |
| EV Load Ratio                | —    |
| T Component Weight           | 0.05 |

SSI Variables Feeding This Report

| SSI VARIABLE                   | VALUE | ESG DISCLOSURE REQUIREMENT          | STATUS |
|--------------------------------|-------|-------------------------------------|--------|
| T1 Transition stress score     | —     | ESRS E1: Transition plan assessment | GAP    |
| DER_ratio Renewable/load ratio | —     | TCFD Transition: Technology risk    | GAP    |
| DER_var Intermittency          | —     | EU Green Deal: Grid flexibility     | GAP    |
| EV_load EV charging ratio      | —     | SDG 7: Clean energy access          | GAP    |

## Data Sources & Vintage

Air Quality Monitoring  
National Environmental Agency

2023 Annual

ISO 9223 Corrosion  
ISO

2012 N/A

### TECHNICAL VALIDITY

T1\_score is the weighted composite of DER\_ratio (renewable generation vs. local demand), DER\_variability (intermittency), and EV\_load\_ratio (electric vehicle charging load as fraction of capacity). DER data sourced from national energy regulators and renewable installation registers. DER\_ratio > 1.0 indicates net export during peak generation — a marker of successful energy transition with associated infrastructure stress.

## 6 Cybersecurity Exposure

NIS2 Directive · ENISA Cybersecurity Index · ESRS G1 Governance

GAP

### WHY THIS REPORT EXISTS

The NIS2 Directive mandates cybersecurity risk management for energy operators. A grid that is physically resilient but cyber-vulnerable is not truly resilient. The SSI R7 modifier combines SCADA assessment, communication architecture analysis, and national-level cyber maturity (ENISA/DESI indices) to produce a substation-level digital vulnerability score.

### SDG Alignment

#### SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.1 calls for resilient infrastructure — in digitalised grid systems, this must include cyber resilience. The R7 modifier (—) assesses digital vulnerability combining SCADA exposure, communication protocol security, and national cyber maturity. SDG 16 (Strong Institutions) also applies: NIS2 compliance is a governance imperative, and the betweenness centrality (—) identifies single-point-of-failure risk in the grid topology.

SDG 9

SDG 16

### Substation Profile

|                            |    |
|----------------------------|----|
| R7 Cyber Modifier          | —  |
| Graph Degree (topology)    | —  |
| BC Percentile (centrality) | —  |
| Is Bridge Node             | No |
| Cluster Coefficient        | —  |

### SSI Variables Feeding This Report

| SSI VARIABLE              | VALUE | ESG DISCLOSURE REQUIREMENT          | STATUS |
|---------------------------|-------|-------------------------------------|--------|
| R7 Cyber modifier         | —     | NIS2: Cybersecurity risk management | GAP    |
| BC Betweenness centrality | —     | ESRS G1: Governance & topology risk | GAP    |
| Degree Graph degree       | —     | Grid resilience: Connectivity       | GAP    |

### Data Sources & Vintage

|                                          |               |
|------------------------------------------|---------------|
| Cybersecurity Index<br>ENISA             | 2024 Biennial |
| DESI Connectivity<br>European Commission | 2024 Annual   |

#### TECHNICAL VALIDITY

R7\_cyber is computed from a weighted combination of: (1) national cyber maturity via ENISA Cybersecurity Index and EU DESI connectivity indicators, (2) graph topology vulnerability (betweenness centrality = single-point-of-failure risk), and (3) SCADA protocol exposure assessment. Substation-level SCADA data is operator-proprietary — the national-level proxy provides a baseline but not asset-specific granularity.

WHY THIS REPORT EXISTS

The SFDR (Sustainable Finance Disclosure Regulation) mandates that financial market participants disclose Principal Adverse Impact (PAI) indicators for infrastructure investments. PAI Table 1 requires quantitative assessment of physical + governance risk at asset level for Article 8/9 fund classification. The SSI v4.2 Re composite (resilience metric per FC v3 §14 subsection 13.7) provides the canonical proxy for Infrastructure PAI: it integrates 6-axis physical risk (C/V/I/E/S/T), 6-modifier degradation chain (R3/R4/R6c/R6d/R6e/R8/R9/R10), and Markov forward-looking trajectory into a single normalised value directly consumable by SFDR reporting.

SDG Alignment

SDG 12 — Responsible Consumption and Production (primary)

Target 12.6 encourages companies to adopt sustainable practices and integrate sustainability information into their reporting cycle. SFDR is the EU regulatory instrument that operationalises this at fund level for infrastructure investors. This substation carries an Re\_norm (normalised resilience composite) of 0.468 and Re\_raw of 1.326, providing the empirical basis for PAI Table 1 Infrastructure indicator disclosure under Delegated Reg (EU) 2022/1288.

SDG 12

SDG 9

SDG 13

Substation Profile

|                              |       |
|------------------------------|-------|
| Re_norm (SFDR PAI composite) | 0.468 |
| Re_raw (pre-normalisation)   | 1.326 |
| R6c Flood modifier           | 1.094 |
| R6d Wildfire modifier        | 1.007 |
| R6e Winter storm modifier    | 1.150 |
| R8 Adaptation modifier       | 0.977 |
| R9 Compound event modifier   | 1.008 |
| R10 Just-transition modifier | 1.081 |

SSI Variables Feeding This Report

| SSI VARIABLE                            | VALUE | ESG DISCLOSURE REQUIREMENT               | STATUS |
|-----------------------------------------|-------|------------------------------------------|--------|
| Re_norm SFDR PAI composite (normalised) | 0.468 | SFDR Art. 4 · PAI Table 1 Infrastructure | READY  |
| Re_raw Pre-normalisation resilience     | 1.326 | FC v3 §14 subsection 13.7                | READY  |
| R6c Flood modifier                      | 1.094 | SFDR PAI: Physical acute hazards         | READY  |
| R6d Wildfire modifier                   | 1.007 | SFDR PAI: Physical acute hazards         | READY  |
| R6e Winter storm modifier               | 1.150 | SFDR PAI: Physical acute hazards         | READY  |
| R8 Adaptation modifier                  | 0.977 | SFDR PAI: Adaptation planning            | READY  |
| R9 Compound event modifier              | 1.008 | SFDR PAI: Compound climate risk          | READY  |
| R10 Just-transition modifier            | 1.081 | SFDR PAI: Social transition risk         | READY  |

Data Sources & Vintage

No dedicated data sources for this report.

TECHNICAL VALIDITY

Re\_norm is the canonical SFDR PAI Infrastructure composite per SSI Index v4.2 methodology (FC v3 §14 subsection 13.7). It integrates the 8 v4.2 modifiers (R6c/R6d/R6e physical acute hazards + R8 adaptation + R9 compound events + R10 just-transition + R7 cyber + R3 social) into a single value normalised against the country-fleet percentile distribution. Convention #56 visibly-honest degradation preserves Re\_norm=0.0 + Re\_raw=1.0 as neutral defaults for net-new substations awaiting the L2/L3/L4 modifier-chain rescore pass per Convention #78 §4bis.4 two-phase workflow. GAP status reflects sites with fewer than 50% of fleet subs carrying non-default modifiers (typical during a v4.23 L1 refresh window before the follow-on rescore).

SFDR Principal Adverse Impact (PAI) Infrastructure v4.2 — Re

7

composite anchor

SFDR Article 7 PAI · ESRS E1-7 climate-resilience disclosure · CSRD double-materiality infrastructure overlay (per FC v3 §14 subsection 13.7)

READY

SFDR Article 7 requires asset-level disclosure of Principal Adverse Impacts on sustainability factors for infrastructure investments. The Re composite (v4.2 bounded resilience aggregate, bounds [0.920, 1.787] normalised to [0, 1]) is the SSI's canonical driver for PAI Infrastructure exposure — it captures the substation's standing on the new v4.2 resilience-modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in a single scalar suitable for SFDR PAI reporting. CSRD double-materiality requires both inside-out (impact on environment) and outside-in (climate-resilience risk to the asset) — Re composite anchors the outside-in dimension for grid infrastructure. This report closes the FC v3 §14 subsection 13.7 R7 SFDR PAI axis on the ESG Radar.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.1 calls for developing quality, reliable, sustainable and resilient infrastructure. The Re composite  $Re\_norm = 0.468$  (moderate exposure) at this substation provides the SFDR PAI Infrastructure metric, anchoring climate-resilience disclosure to a single auditable scalar.  $Re\_raw = 1.3256$  with methodological bounds [0.920, 1.787] derived from the v4.2 modifier family product range. Lower  $Re\_norm \rightarrow$  lower PAI Infrastructure exposure  $\rightarrow$  stronger SFDR Article 7 alignment.

SDG 9

SDG 13

SDG 11

Substation Profile (Re composite)

|                             |                           |
|-----------------------------|---------------------------|
| Substation ID               | PL_v43_ff22713eca7d       |
| Re_raw (unbounded)          | 1.3256                    |
| Re_norm (bounded [0, 1])    | 0.468                     |
| Tier classification         | MODERATE                  |
| Methodology bounds          | [0.920, 1.787]            |
| Polish fleet observed range | 1.118 $\rightarrow$ 1.144 |
| Coverage                    | 2,248 substations         |

SSI Variables Feeding This Report

| SSI VARIABLE                                                    | VALUE  | ESG DISCLOSURE REQUIREMENT                    | STATUS |
|-----------------------------------------------------------------|--------|-----------------------------------------------|--------|
| R6c Flood (ISPRA PGRA, ADDITIVE +0.00..+0.30)                   | +1.094 | SFDR PAI 4 (climate physical risk)            | READY  |
| R6d Wildfire (EFFIS + FIRMS + SIAB, MULT)                       | 1.007  | SFDR PAI 4 (climate physical risk)            | READY  |
| R6e Winter / Ice (ERA5 Bourgouin p99, MULT)                     | 1.150  | SFDR PAI 4 (climate physical risk)            | READY  |
| R8 Adaptive Capacity (URE + ME + DESI Poland)                   | 0.977  | ESRS E1-3 Adaptation actions                  | READY  |
| R9 Compound Hazard (ISPRA RNDA, STEP)                           | 1.008  | ESRS E1-9 Anticipated financial effects       | READY  |
| R10 Energy Justice (EU-SILC + Poland SILC + national regulator) | 1.081  | SFDR PAI 14 (board diversity proxy) + ESRS S3 | READY  |
| Re_norm aggregate (bounds [0.920, 1.787])                       | 0.468  | SFDR Art. 7 PAI Infrastructure anchor         | READY  |

### Data Sources & Vintage

|                                                                     |                               |
|---------------------------------------------------------------------|-------------------------------|
| ISPRA IdroGEO PGRA                                                  | 2024 (annual)                 |
| Copernicus EFFIS + NASA FIRMS                                       | Q4 2025 (active + historical) |
| ISPRA SIAB national-tier burned-area                                | CY2024                        |
| ECMWF ERA5 Bourguoin freezing-rain p99                              | 1991-2020 reanalysis          |
| URE quality regulation indicators + PSE+DSO smart-meter penetration | 2024 annual                   |
| ENEA Adaptation Index                                               | 2024                          |
| DESI Digital Economy index                                          | 2024                          |
| ISPRA RNDA compound-hazard cross-product                            | CY2024                        |
| EU-SILC ILC_MDES01/O3                                               | 2024 wave                     |
| Eurostat EU-SILC + GUS EU-SILC PL (Polish energy-poverty proxy)     | 2024                          |
| URE regional SAIDI                                                  | CY2024                        |

### TECHNICAL VALIDITY

The Re composite is the bounded aggregate  $Re\_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$  with normalisation  $Re\_norm = clip((Re\_raw - 0.920) / (1.787 - 0.920), 0, 1)$ . Bounds [0.920, 1.787] derived from the multiplicative band of the 5 multiplicative v4.2 modifiers at tier extremes plus the R6c additive cliff range. Polish fleet observed range [0.974, 1.604] across 2,248 substations. SFDR PAI Infrastructure proxy validated against TCFD physical-risk framework and ESRS E1-9 anticipated-financial-effects disclosure. See FC v3 §14 subsection 13.7 for the R7 SFDR PAI axis derivation.

## C Audit Trail & Methodology Note

Traceability for CSRD Article 29a limited assurance

| ITEM                       | VALUE                                                                                                                                                                |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scoring Engine             | SSI Index <b>v4.2</b> — Gaussian copula Monte Carlo (10,000 iterations) · 11 modifiers (5 baseline + 6 v4.2 resilience family) · Re composite bounded [0.920, 1.787] |
| Weight Architecture        | C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05                                                                                                                       |
| Modifiers Applied          | R6 seismic: 1.007, R6c flood: 1.094, R6d wildfire: 1.007, R6e winter: 1.150, R8 adapt: 0.977, R9 compound: 1.008, R10 just: 1.081                                    |
| Degradation Model          | Markov 5-state, CIGRE TB 761 calibration                                                                                                                             |
| Thermal Model              | IEEE C57.91 transformer loading curves                                                                                                                               |
| Report Date                | 17 March 2026                                                                                                                                                        |
| Reporting Period           | Calendar year 2025                                                                                                                                                   |
| Refresh Cadence            | Annual (data ingestion → scoring engine → display)                                                                                                                   |
| Substation                 | PL523 · Station v43_ff22713eca7d (PL_v43_ff22713eca7d)                                                                                                               |
| Data Assurance Level       | All inputs from institutional open-source with citable vintage                                                                                                       |
| <b>Re composite (v4.2)</b> | Re_raw = 1.3256 · Re_norm = <b>0.468</b> · <b>MODERATE</b> exposure · bounds [0.920, 1.787]                                                                          |

D

## Re composite — bounded resilience aggregate (v4.2)

Resilience composite anchoring CSRD/ESRS climate-resilience disclosure

The Re composite (v4.2) is a bounded resilience aggregate capturing the new R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just modifier family in a single scalar suitable for CSRD/ESRS climate-resilience disclosure:

$$\text{Re\_raw} = (\text{R6d} \times \text{R6e} \times \text{R8} \times \text{R9} \times \text{R10}) + (\text{R6c} - 1.00)$$

Bounds: **[0.920, 1.787]**. Re\_raw < 1.000 when R8\_adapt < 1.00 dominates other near-neutral factors. Polish fleet observed range across 2,248 substations: the methodology-bounded extremes [0.920, 1.787] — low-end values arise at interior low-exposure substations with R8\_adapt < 1.00 dominating near-neutral other modifiers, high-end values at coastal compound-exposure substations where R6c flood + R9 compound stack multiplicatively. v4.2 explicitly retires the pre-Path-B "winter exposure is a real climate burden" reading per Convention #6 §2.3.1 (Bourgouin freezing-rain p99 tail-event amplifier per §6 design — winter exposure signal correctly lives in v4.0.2 I1 Snow/Ice metric, where it methodologically belongs). See [W1-W10 FC v2](#) for per-substation worked examples + W4 Climate resilience baselines.

SSI Index v4.0.2 · Ikenga Enhanced Infrastructure Intelligence · Annual ESG Disclosure

Report generated 21 May 2026 · All data sources are open-access institutional

### Foundation contact

The SSI Index is a non-profit foundation project, please mail to [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu) for more details.